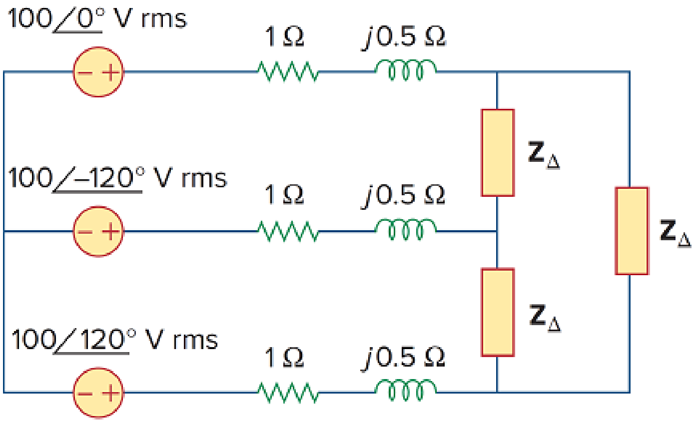
Chapter 12, Problem 40P

Problem

For the three-phase circuit in Fig. 12.59, find the average power absorbed by the delta-connected load with **Z**Δ = 21 + *j*24 Ω.

Figure 12.59



Step-by-step solution

Step 1 of 4

Refer to Figure 12.59 in the textbook.

Transform the $\Delta-$ network into $Y-$network.

Consider the load impedance $Z_\Delta = 21 + j24\ \Omega$.

Determine the equivalent load impedance of $Y-$network Z_Y.

Consider the equation for the Z_Y.

$$\begin{aligned} \mathbf{Z}_Y &= \frac{\mathbf{Z}_\Delta}{3} \\ &= \frac{(21 + j24)}{3} \\ &= 7 + j8\ \Omega \end{aligned}$$

Step 2 of 4

Determine the value of $\mathbf{I}_a$.

Consider the equation for $\mathbf{I}_a$.

$$\mathbf{I}_a = \frac{\mathbf{V}_{rms}}{\mathbf{Z}_A} \quad \ldots\ldots (1)$$

Determine the value of $\mathbf{Z}_A$.

$$\begin{aligned} \mathbf{Z}_A &= (1 + j0.5) + (7 + j8) \\ &= 8 + j8.5\ \Omega \end{aligned}$$

Replace $100\angle 0°$ for $\mathbf{V}_{rms}$ and $(8 + j8.5)\ \Omega$ for $\mathbf{Z}_A$ in equation 1.

$$\begin{aligned} \mathbf{I}_a &= \frac{100\angle 0°}{(8 + j8.5)} \\ &= 5.87 - j6.23\ \text{A} \end{aligned}$$

Determine the complex power.

$\mathbf{S} = 3|\mathbf{I}_a|^2(\mathbf{Z}_Y)$ ...... (2)

Replace $7 + j8\ \Omega$ for $\mathbf{Z}_Y$, and $5.87 - j6.23\ \text{A}$ for $\mathbf{I}_a$ in equation 2.

$$\begin{aligned}
\mathbf{S} &= 3|\mathbf{I}_a|^2(\mathbf{Z}_Y) \\
&= 3(8.567)^2(7 + j8\ \Omega) \\
&= 220.18(7 + j8) \\
&= 1541.26 + j1761.44
\end{aligned}$$

Determine the average power absorbed by load.

Consider the equation for average power.

$P = \text{Re}(\mathbf{S})$ ...... (3)

Consider the real part of complex power.

$$\begin{aligned}
P &= \text{Re}(\mathbf{S}) \\
&= 1541.26\ \text{W}
\end{aligned}$$

Therefore, the average power absorbed by the load is $\boxed{1541.26\ \text{W}}$.